

Report Explaining the Architecture:

This report details the architectural design for the Grocery Sales Forecasting solution. The focus of this implementation is to provide a robust, leakage-proof pipeline that prioritizes validation discipline and domain-specific feature engineering.

1. Forecasting Architecture Overview

The pipeline is designed with a **decoupled configuration pattern**, separating core data engineering logic from model hyperparameter definitions. This modularity ensures that the feature engineering pipeline acts as the "Single Source of Truth" across all experimental branches.

- **Modular ETL Engine:** The pipeline ingests raw streams, performing multi-table joins on store metadata and macroeconomic oil indices. To address data gaps (e.g., weekend/holiday closures), we apply continuous chronological grid propagation (ffill/bfill).
- **Leakage-Proof Validation Strategy:** The system utilizes a **16-day blind-spot lag engine**. To prevent data leakage—where information from the future is inadvertently included in training—all historical lags (16–28 days) and rolling statistics are locked behind a strict 16-day cutoff boundary. This ensures the model learns only from information available to a real-world operator on the date of prediction.

2. Feature Strategy & Domain Engineering

The model is enriched with domain-specific features that capture retail demand shocks:

- **Promotional Momentum:** Beyond static flags, we engineered rolling sums (7-day/14-day) and "forward-leads" for promotional schedules. Since retailers determine promos in advance, these leads provide the model with "foresight" without violating leakage constraints.
- **Demand Velocity:** We introduced target acceleration features—specifically the difference between historical lag windows (e.g., lag 16 – lag 17). This allows the model to capture the **slope** of demand trends, identifying whether a product is currently accelerating or crashing in volume.
- **Regional & Macro Primitives:** The model incorporates bi-weekly payday flags and an earthquake recovery window, which were empirically validated to reduce noise during extreme demand volatility.

3. Metric Design & Ensemble Philosophy

The pipeline optimizes for **RMSLE (Root Mean Squared Logarithmic Error)** rather than raw RMSE.

- **Log-Space Normalization:** By training in log-space, we successfully neutralized "Big Box Bias"—the tendency for models to over-prioritize flagship stores at the expense of smaller, low-volume families. This approach forces the tree models to treat percentage errors symmetrically across all 1,782 time-series combinations.

- **Dual-Engine Ensemble:** To maximize robustness, the production architecture employs a weighted blend (85% LightGBM / 15% XGBoost). LightGBM's leaf-wise growth captures granular, hyper-localized product trends, while XGBoost's level-wise growth provides structural stability for low-volume categories. Fusing these models mitigates the systematic variance of either individual paradigm.